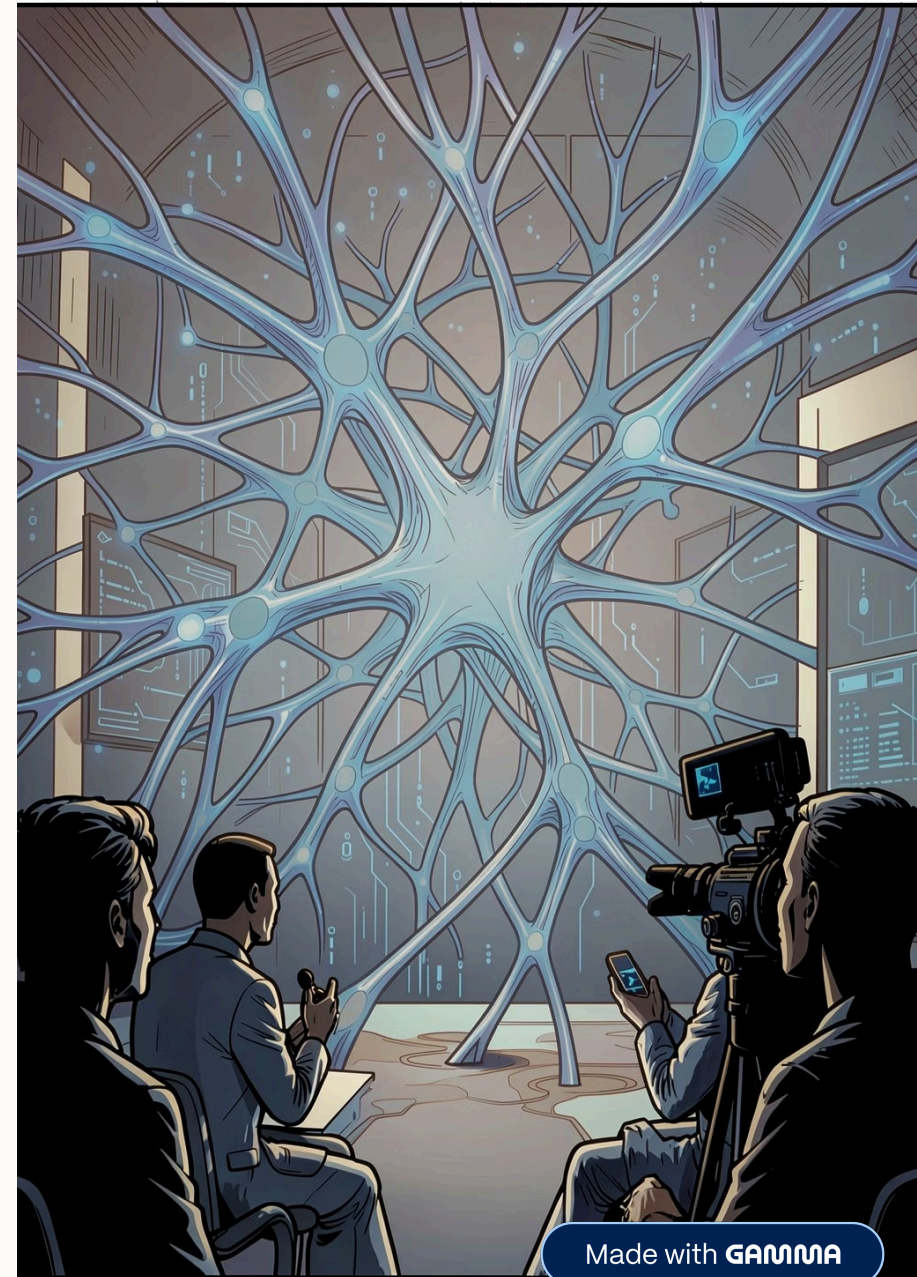


AI-Powered Interview Preparation Platform

A personalized, intelligent system that helps students prepare for technical interviews using LLMs, RAG, and skill gap analysis.

SRIVALLI KUKKADAPU-23EG112C54(IT/C)



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Introduction & Abstract

This AI-powered platform helps students prepare for technical interviews in a simple and personalized way. It uses LLMs, RAG, and skill gap analysis to find what each student needs to improve. Then it provides focused practice and learning content based on their progress. This makes interview preparation more effective and easier to follow.





The Problem with Traditional Interview Prep

No Personalization

Generic study materials fail to address individual skill gaps or target specific job requirements.

Lack of Guidance

Students receive no structured roadmap or real-time feedback to improve their responses.

Inefficient Practice

Without mock interviews and answer evaluation, candidates struggle to identify and fix weaknesses before the real interview.

Solution Overview



An end-to-end AI platform that analyzes your resume and target job description, then generates a fully personalized interview preparation experience.

LLM-Powered

Groq for fast, accurate inference

RAG-Based

LangChain + ChromaDB for context-aware responses

Personalized

Tailored to your resume and JD

Key Features



Resume & JD Parsing

Automatically extracts skills and requirements from uploaded documents.



Answer Evaluation

Provides structured feedback on accuracy, clarity, and completeness.



Skill Gap Detection

Identifies missing competencies by comparing resume against job description.



Study Roadmap

Generates a prioritized learning plan to close identified skill gaps.

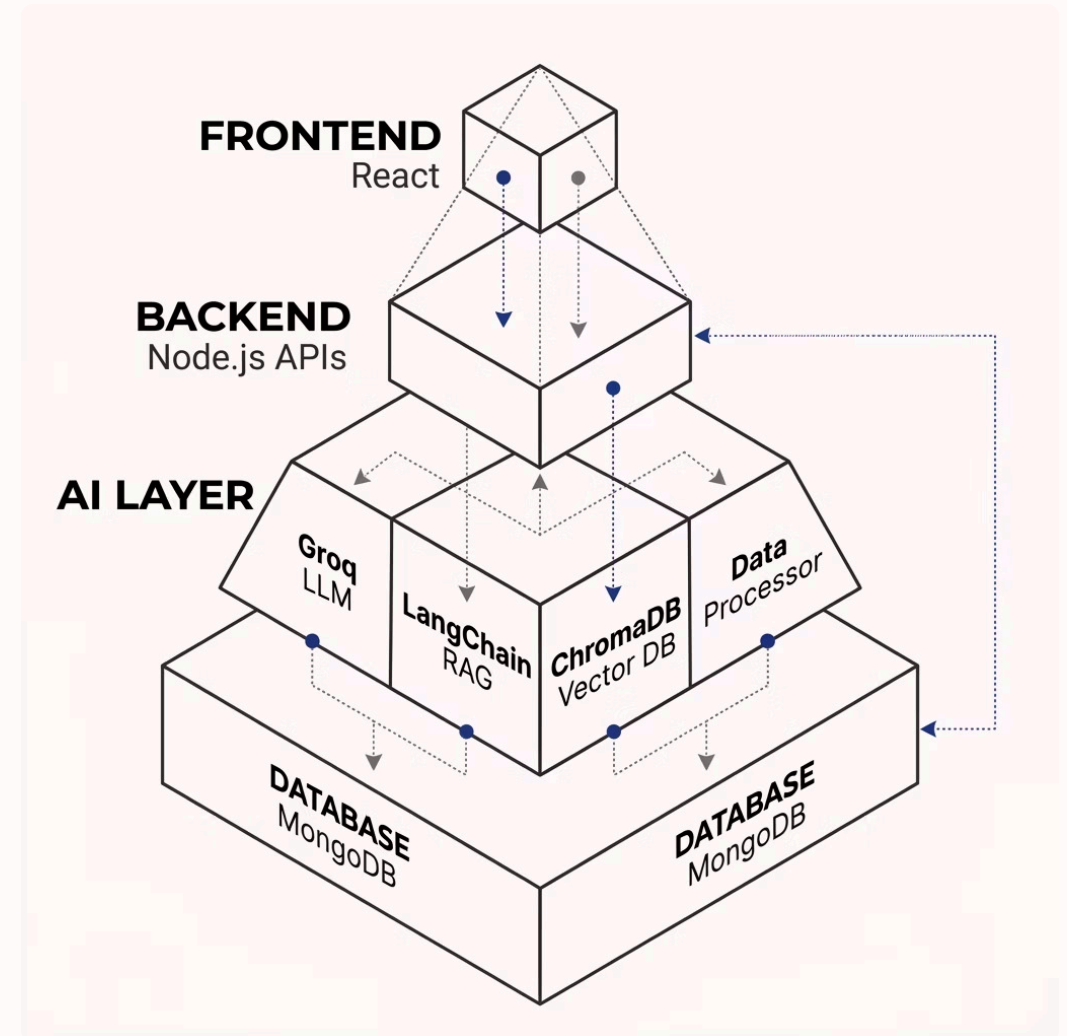


Mock Interviews

Simulates real interview scenarios with AI-generated, role-specific questions.

System Architecture

The platform uses a modern, decoupled architecture ensuring scalability, fast response times, and reliable AI inference through Groq and LangChain.



→ Frontend

React — responsive UI for resume upload, interviews, and dashboards

→ AI Layer

Groq LLM + LangChain RAG + ChromaDB vector store

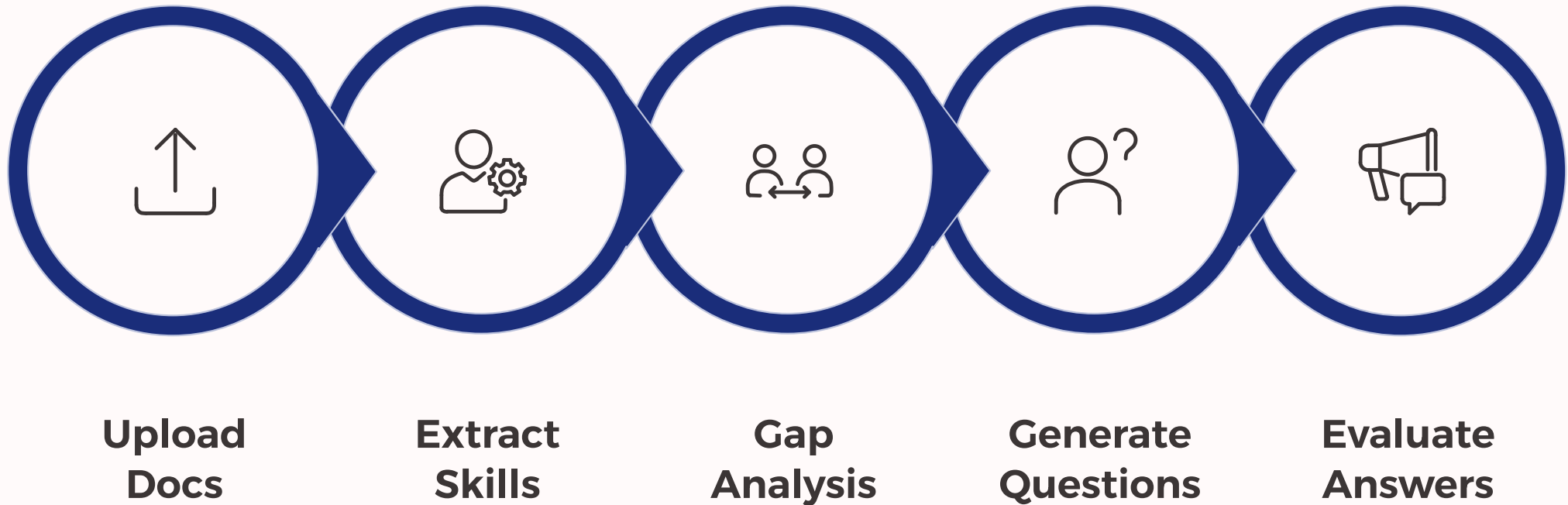
→ Backend

Node.js APIs — handles business logic, authentication, and data flow

→ Database

MongoDB — stores user data, sessions, and evaluation results

End-to-End Workflow



From upload to feedback in seconds — the pipeline automates every step of interview preparation, delivering a fully personalized experience without manual effort.

RAG & AI Implementation

How RAG Reduces Hallucination

Retrieval-Augmented Generation (RAG) grounds the LLM's responses in verified, domain-specific content stored in ChromaDB — dramatically reducing inaccurate or fabricated answers.



Retrieve

Query ChromaDB for relevant context chunks

Augment

Inject context into the LLM prompt

Generate

Groq produces accurate, grounded responses

Evaluation Metrics

95%

Answer Accuracy

Measured against ground-truth reference answers

<2s

Response Latency

Average inference time via Groq API

4.7/5

User Feedback Score

Average rating from beta users

90%

Relevance Rate

Questions matched to target job description

Metrics are tracked continuously to ensure the platform delivers high-quality, reliable interview preparation at scale.

Security & Deployment

Security Features

- **Authentication** — Secure user login with session management
- **Input Validation** — Sanitizes all user-submitted data
- **API Protection** — Rate limiting and access controls

Deployment Stack

Frontend

Vercel — zero-config,
edge-ready deployment

Backend

Cloud-hosted Node.js
with environment-
configured secrets

Database

Managed MongoDB + ChromaDB cloud instance

Impact & What's Next

Today

A fully functional platform delivering personalized mock interviews, skill gap analysis, and AI-powered feedback — helping students land their dream roles.

Next

Multi-language support, video interview analysis, real-time collaboration features, and integration with LinkedIn and job boards for seamless preparation.



Conclusion

Transformative Summary

This AI-powered platform turns interview preparation into a personalized, data-driven experience — helping students practice smarter, improve faster, and enter interviews with confidence.

Key Takeaway

By combining mock interviews, skill gap analysis, and actionable feedback, the platform creates a clear path from preparation to placement.

Call to Action

Prepare with purpose. Build confidence. Land the role you've been working toward.

Thank You

Thank you for your time and attention. I truly appreciate the opportunity to share this with you and hope it was both useful and inspiring.

Srivalli KUKKADAPU-23EG112C54(IT/C)

Questions?

Let's connect and continue the conversation.